

Non-parametric regression II

Exercise 1

As in HW09, we are interested in the `temperature.txt` dataset.

- Fit a GAM model to describe the behavior of January temperature with respect to the longitude.
- Code a cross-validation procedure to select the best bandwidth parameter, according to the GCV criterion. Then, plot the curve of the criterion vs. the value of the parameter.
- Fit three different models with varying smoothing parameter and the optimal one. Plot the corresponding curve along with each other. Does the optimal smoothing parameter yield a sensibly better curve than the other values?

Exercise 2

Let us consider a univariate nonparametric model. As shown during the lesson (*c.f.* slide 271), the basis functions allows the linear representation:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \epsilon.$$

In order to control the wiggleness, the parameters $\boldsymbol{\beta}$ are estimated *via* minimization of a penalized cost function:

$$\mathcal{C}(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda\boldsymbol{\beta}^T \mathbf{S}\boldsymbol{\beta}.$$

In this exercise, we reparametrize the model in order to understand better some of its features. The so-called *natural parametrization* is described in the course (slide 292).

- Following the course, define $\mathbf{P} = \mathbf{U}^T \mathbf{R}$, $\boldsymbol{\beta}' = \mathbf{P}\boldsymbol{\beta}$ and $\mathbf{X}' = \mathbf{Q}\mathbf{U}$ (allowing $\mathbf{X} = \mathbf{X}'\mathbf{P}$), and express the cost function $\mathcal{C}(\boldsymbol{\beta}')$ under this new specification.
- Provide the estimator $\hat{\boldsymbol{\beta}'}$ of $\boldsymbol{\beta}'$ minimizing $\mathcal{C}(\boldsymbol{\beta}')$.
- Deduce the linear smoother $\mathbf{A}$ such that $\hat{\mathbf{y}} = \mathbf{A}\mathbf{y}$. Compute $\text{tr}(\mathbf{A})$ and find an interpretation for this quantity.
- Explain what happens to $\mathbf{A}$ as λ increases.

Exercise 3

The `mcycle` dataset (package `MASS`) records a series of measurements of head acceleration in a simulated motorcycle accident, used to test crash helmets. The variables are `times` in milliseconds after impact, and `accel` in G.

- a) Fit a GAM model to capture the dependence of `accel` on `times`. Use `gam.check()` to fix the dimension of the basis k .
- b) In Exercise 2, we introduced the linear smoother $\mathbf{A}$, such that $\hat{y} = \mathbf{A}y$. Here we compute the same matrix using a convenient numerical device. First, note that $\mathbf{A}_j = \mathbf{A}\mathbf{I}_j$, where $\mathbf{A}_j$ is the j th column of $\mathbf{A}$ and $\mathbf{I}_j$ is the j th column of the identity matrix. Thus, given a model, you can use the columns of the identity matrix in place of the response variable to generate the columns of $\mathbf{A}$: `A_j <- fitted(gam(I_j ~ s(x), ...))`. Making use of this method, compute the influence matrix corresponding to your model in point [a](#).
- c) What values do the rows of $\mathbf{A}$ sum to? How is it related to the intercept of your model?
- d) The linear representation of the smoother shows that our model replaces each values y_i by a weighted sum of the neighbouring y_i . Plot the weights used to compute the 65th predicted value, with `times` in abscissa. This is known as the "equivalent kernel". Interpret the role of this kernel, and describe its specific aspects.
- e) Plot all the equivalent kernels (for rows $i = 1, \dots, n$) on the same frame. Why do their peak heights vary?
- f) Finally, move the smoothing parameter around the GCV selected value. What happens to the equivalent kernel for the 65th observation? What can you conclude?